

TECHNICAL REPORT

Security Features and Governance Practices in Zoho Creator

Experiment 12

Name: MAITREYEE DD

Register No.: 241401050

Table of Contents

1. Aim
2. Why Security and Governance Matter
3. Example App: Hostel Room and Complaint Management
4. Security Features Used
5. Governance and Admin Practices
6. Step-by-Step Setup
7. Testing the Setup
8. Limitations
9. Tips
10. Conclusion
11. References

1. Aim

The aim of this report is to show how security features and good admin habits can be applied to an app built in Zoho Creator. A college hostel app is used as the example. The report explains who can log in, who can see what, and how changes and users are looked after over time.

2. Why Security and Governance Matter

Zoho Creator makes it easy to build and publish an app quickly. But an app that stores student names, phone numbers and complaints must protect that information. If everyone can see everything, privacy is lost. If anyone can edit records without a trace, trust is lost.

Two simple terms are used in this report:

- **Security:** Making sure only the right people can enter the app and see only what they should.
- **Governance:** Managing the app properly over time: adding and removing users, testing changes, and keeping a record of actions.

3. Example App: Hostel Room and Complaint Management

3.1 About the App

The app helps a college hostel handle two things: room allotment and student complaints. It has these parts:

- **Room_Request_Form:** Student name, roll number, year, department, phone number, parent phone number and preferred room type.
- **Room_Master_Form:** Room number, block, floor, bed count and status (free or full).
- **Complaint_Form:** Complaint type (water, electricity, cleaning, food, other), description, photo and status.
- **Hostel_Dashboard:** A page showing rooms filled, pending complaints and complaints by type.

Phone numbers, parent details and complaints about people are private. That is why access must be planned with care.

3.2 Who Uses the App

User Group	Who	What They Need
Admin	IT staff / app owner	Manage users, settings and releases
Hostel Warden	Head of the hostel	See all requests and complaints, approve rooms
Block Supervisor	Staff for one block	See only their own block's data
Auditor	College office	View reports and audit trail only
Student	Hostel resident	Submit requests, see only own records

Table 1: User groups

4. Security Features Used

4.1 Sign-In Protection

Zoho Creator uses security policies (through Zoho Directory) to protect login. These are available on paid plans, so they may not work on a free trial. The settings chosen for this app are:

- **Password rules:** Strong passwords with a minimum length and a mix of letters, numbers and symbols.
- **Multi-factor authentication (MFA):** A second check (like an OTP) at login, first for Admin and Warden accounts.
- **IP limits:** Staff can log in only from the college network or VPN.
- **Session limits:** Idle sessions end after a short time, so an unlocked lab computer is not a risk.

If the college already has student email accounts, single sign-on (SSO) can be used so students log in with their existing account. When a student leaves the college and the account is switched off, hostel app access ends too.

4.2 Permissions

Permissions decide what a logged-in person can do with each form, report, page and field. The idea followed is least privilege: give only the access that is truly needed. A suggested design is shown below.

Part of App	Warden	Supervisor	Auditor	Student
Room_Request_Form (submit)	Yes	No	No	Yes
Request report	All	Own block	View only	Own only
Complaint report	All	Own block	View only	Own only
Phone / parent details	View, edit	View	View	Own only
Approve room	Yes	No	No	No
Delete records	No	No	No	No
Export data	Yes	No	Yes	No
Hostel_Dashboard	Yes	Block view	Yes	No

Table 2: Suggested permissions

Deleting is switched off for everyone so no record disappears quietly. Mistakes are fixed by editing, and the audit trail records every edit.

4.3 Roles and Role Hierarchy

Roles copy the real structure of the hostel into the app. The Warden sits at the top, Block Supervisors below, and Students at the bottom. With role hierarchy on, the Warden can see the records of everyone below, while a Supervisor sees only their own block. Students cannot see each other's data. Peer sharing is kept off because supervisors of different blocks do not need each other's records.

4.4 Data Sharing Rules

Data sharing rules handle exceptions. For example, the Food Committee coordinator could be given only the Complaint report filtered to food complaints. Zoho says these rules override the role

hierarchy and they only add access, so each rule should have a reason, an owner and an end date. A forgotten rule is a common cause of accidental data exposure.

4.5 Marking Personal Information

Creator lets the admin mark fields as personal information. Phone number, parent phone number and address were marked this way, so new roles do not see them by default and get access only when truly needed.

5. Governance and Admin Practices

5.1 Development, Stage and Production

Creator provides three environments so changes are tested safely.

Environment	Use
Development	Where the app is built and edited
Stage	Where changes are tested with demo users
Production	The live app used by real students and staff

Table 3: Environments

A new role or permission is first tested in Stage, so a mistake never exposes real student data. Data in each environment is kept separate.

5.2 Audit Trail

The audit trail records who did what, when, and on which data. In this app it can answer questions such as who changed a room allotment or who closed a complaint. The record is only useful if someone reads it, so a named person (the Auditor) checks it every week.

5.3 Backups and Updates

Zoho takes automatic backups and applies updates and security fixes. The admin should still know how to ask for a restore and should check key forms and reports after every update.

5.4 Looking After Users

- **New user:** Give the role that matches the job and nothing extra.
- **Role change:** Change the existing role instead of piling on new ones.
- **Person leaves:** Switch off the account on the same day. At the end of each academic year, remove students who have left.
- **Regular check:** Review all users, roles and sharing rules every three months.

6. Step-by-Step Setup

Step	Action	Where in Creator
1	List user groups and what each needs	Planning notes
2	Set password, MFA, IP and session rules	Security policies (Zoho Directory)
3	Connect college accounts for SSO, if available	SAML / directory sync

Step	Action	Where in Creator
4	Create roles and switch on hierarchy	Users and Roles
5	Set permissions for forms, reports and fields	Permissions
6	Mark personal fields as sensitive	Permission settings
7	Add sharing rules only for exceptions	Data sharing rules
8	Test with demo users in Stage	Environments
9	Publish to Production	Environments
10	Check audit trail weekly, review access every 3 months	Admin routine

Table 4: Setup plan

7. Testing the Setup

After setup, the app should be tested using one demo account for each role. The table lists the checks and the result expected.

Test	Expected Result
Student opens another student's complaint	Not visible
Block Supervisor opens another block's data	Not visible
Warden opens all requests	All visible
Auditor tries to edit a record	Blocked, view only
Anyone tries to delete a record	Not allowed
Login from an unapproved network (staff)	Blocked
Edit made to a room allotment	Shown in the audit trail

Table 5: Test checklist

8. Limitations

- **Plan limits:** Security policies need a paid plan, so MFA and IP limits may be missing on a trial account.
- **Testing gaps:** Some items, such as approval workflows and sharing with specific users, cannot be fully tested in Development or Stage, so they need extra care at release.
- **Overlapping rules:** Permissions, role hierarchy and sharing rules affect each other. A wrong mix can give too much access, so each change must be tested.
- **Vendor dependence:** Backups and patches are handled by Zoho. For audit proof, the admin must rely on Zoho's published documents.
- **People mistakes:** No setting can stop someone from sharing their password. Short training is also needed.

9. Tips

- Decide roles and private fields before building forms.

- Start with minimum access and add more only when asked.
- Use MFA for all admins and keep the number of admins small.
- Never test in Production.
- Write down the reason, owner and end date of every exception.
- Read the audit trail on a fixed schedule.
- Remove old accounts quickly.

10. Conclusion

Zoho Creator offers layered tools to keep an app safe and well managed. Login rules protect the front door, permissions and roles control what each person sees, and environments, audit trails and backups keep the app organised. In the hostel app example, these tools work best when planned together: decide roles first, give least access, test before release and review often. Security is not a one-time job but a regular habit.

11. References

- Zoho Creator Help: Understanding permissions. Zoho Corporation. help.zoho.com
- Zoho Creator Help: Roles and role hierarchy. Zoho Corporation. help.zoho.com
- Zoho Creator Help: Security policies and Zoho Directory. Zoho Corporation. help.zoho.com
- Zoho Creator Help: Environments and their limitations.
zoho.com/creator/help/deploy/environments
- Zoho Creator Evaluation Guide: Govern. zoho.com/creator/evaluation-guide/govern.html
- Features and plans may change, so recheck current Zoho documentation.